

Week 11

Marketing and Data Science



Goal of this Lecture



Applying What You've Learnt So Far in Real-World Business Scenes

1. The Key Concepts of Data Science

Week 1

1. Why learn data science?

2. Python programming

Week 1, 2

2. Numpy

3. Pandas

4. Matplotlib

3. Basics of machine learning

Week 2, 3

5. Supervised Learning

6. Model Evaluation and Feature Engineering

7. SQL

8. Unsupervised Learning

4. Applying Data Science in Real-World Business Scenes

Week 3

9. Marketing and Data Science

10. Live Lecture: Student Entrepreneurship Using Data Science

01

Introduction to Marketing & Data Science

- [Why Marketing?](#)
- Fundamentals of [Marketing](#), [Marketing Strategy](#), [Data Strategy](#), and Key Technological Terms
- Case Studies

02

Hands-on Marketing Analysis (Designed for Self-study)

- Demonstration using Jupyter Notebook
- Data exploration, decile analysis, RFM analysis, etc.
- Recommendation engine, causal inference, etc.

03

The Next Steps

- Theoretical to [Practical Knowledge](#)
- [Starting your own business](#)



Introduction to Marketing & Data Science

01

Why Marketing?



Marketing as an ideal starting point for real-world application

① Clear, Measurable Goals (KPIs & KGIs):

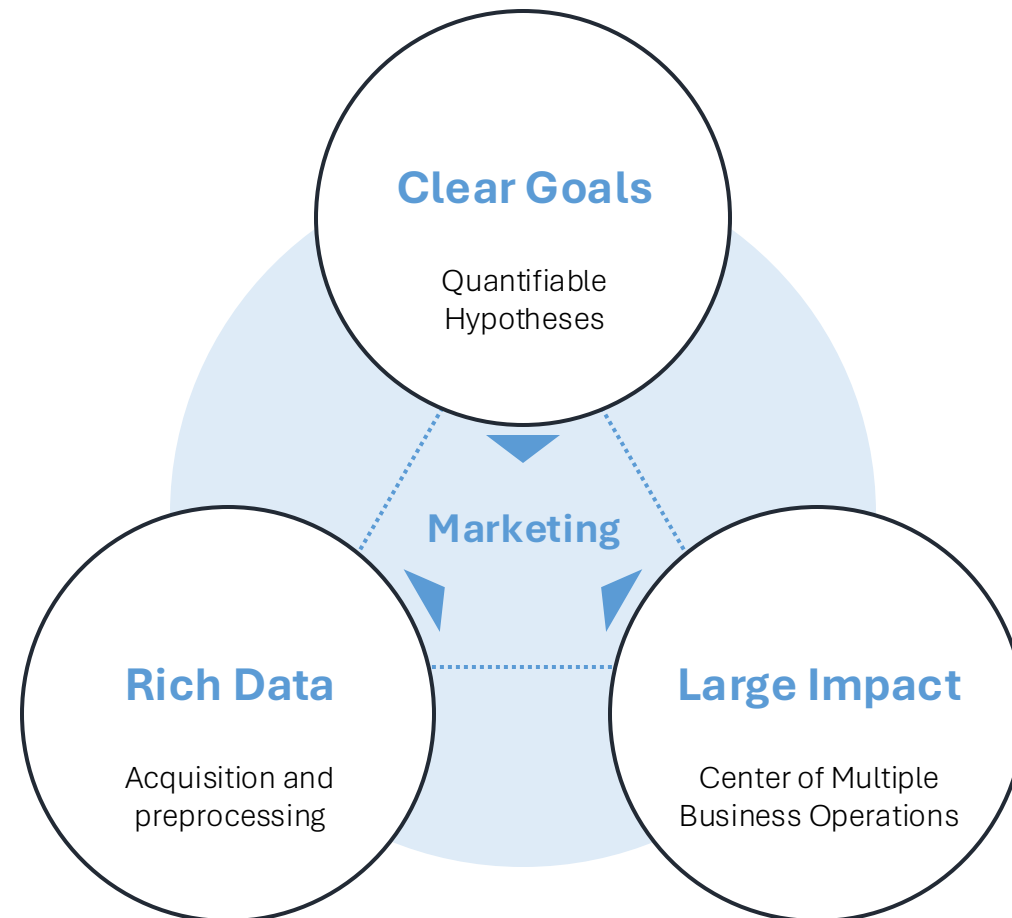
- **Well-defined, measurable goals** (increasing sales, customer retention, reducing churn, and improving customer engagement etc.)
- These goals naturally translate into **quantifiable hypotheses**

② Rich Availability of Data:

- Inherently produces **vast quantities of data**, such as:
 - Behavioral, Transactional, Demographic data, etc.
- An ideal domain for practicing data acquisition and preprocessing skills.

③ Center of many business operations:

- Often serves as a hub connecting various departments such as R&D, Sales, Customer Support, and Supply Chain Management
- **Hence holding a large impact in business scenes**



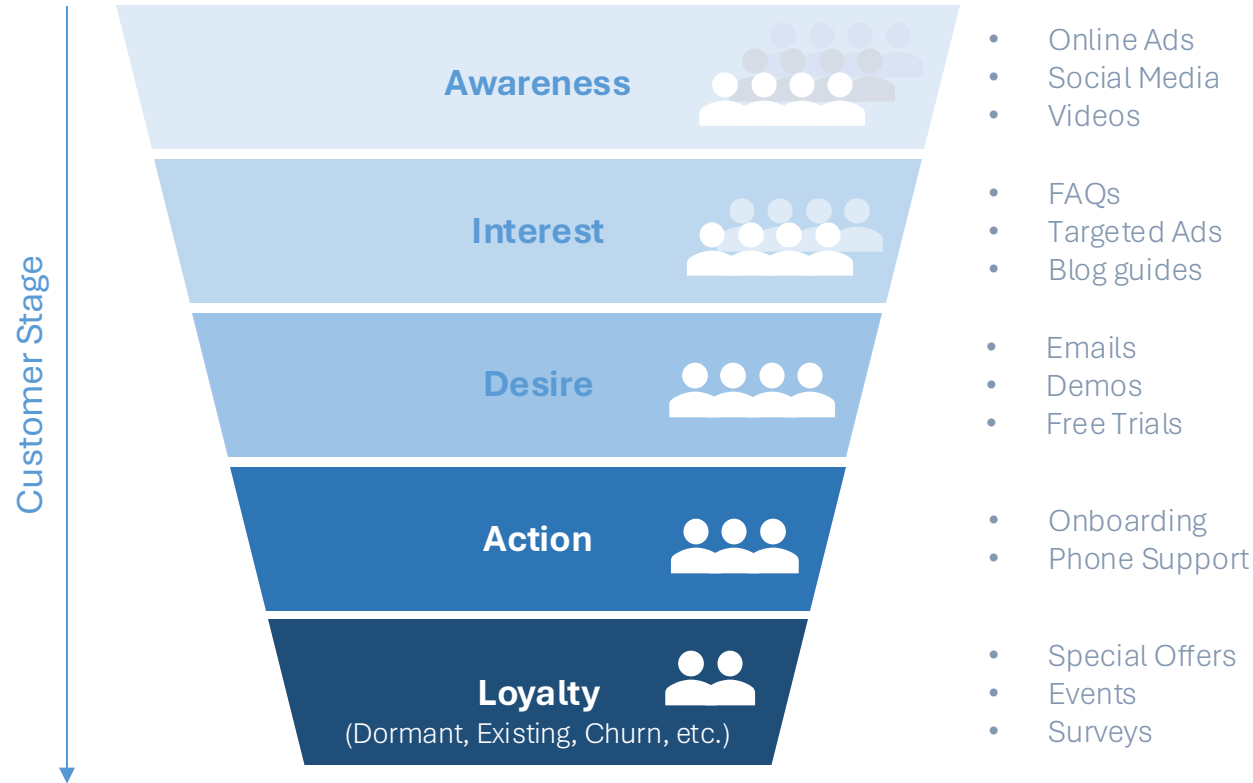
The Fundamentals



Fundamentals of Marketing



Marketing is the process of creating strategies and systems to sell products or services **profitably and efficiently**.



Key Marketing Questions:

- **Which stage** is this specific customer in?
- How can we efficiently **attract new customers**?
- What encourages existing customers to **spend more**?
- How do we **re-engage dormant customers**?
- Which stage has an issue in **conversion rate**?
- Can we **predict and prevent customer churn**?

Customer Stage Varies Among Customers and Requires Different Approaches.

⇒ Data is Used to **Identify, Personalize, and Evaluate** Marketing Strategies

Data Usage & Strategies in Familiar Services

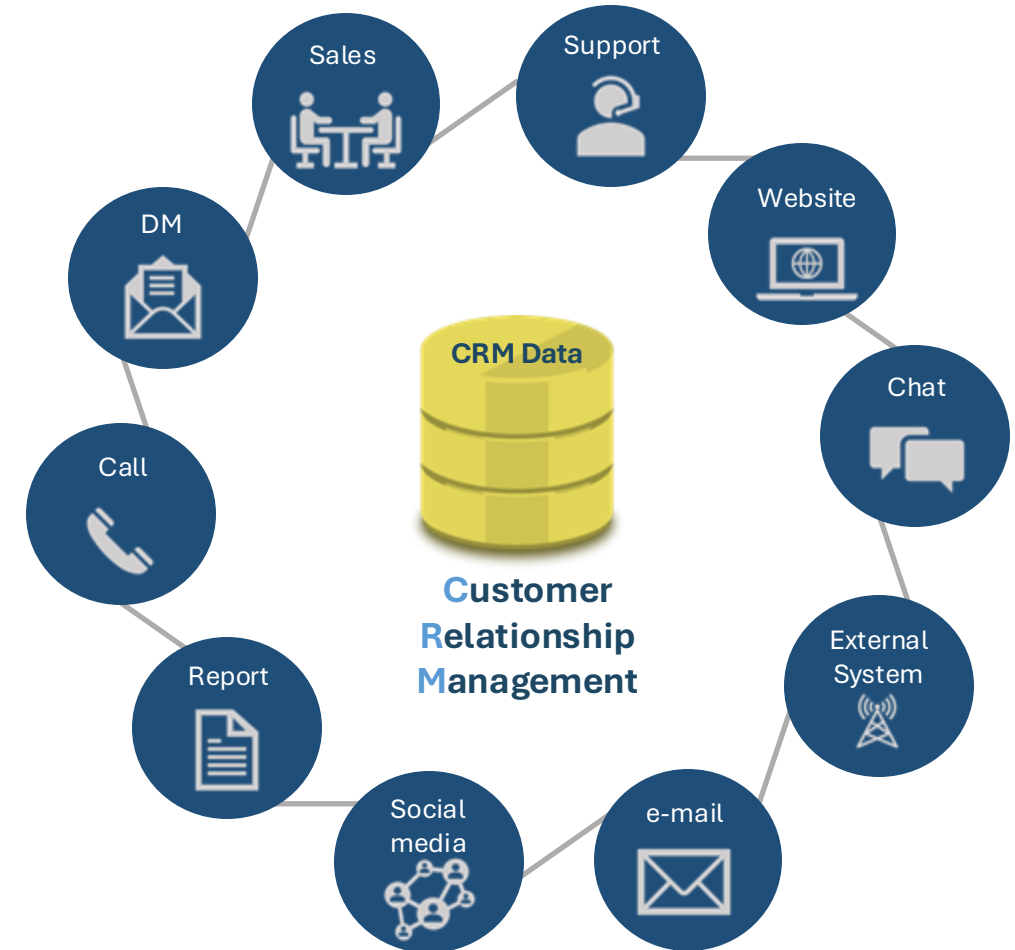


How Marketing Strategies are Implemented in our Daily Lives



Data that is Being Used

- **Browsing and Interaction Data:** User clicks, session duration, content viewed, user engagement.
- **Transactional and Purchase Data:** Purchase history, transaction frequency, order value.
- **Demographic and Profile Data:** Age, location, preferences, and customer segment information.
- **Behavioral and Temporal Data:** Frequency, recency, patterns in customer behavior over time.
- **Feedback and Sentiment Data:** Customer reviews, ratings, and survey responses.



Data Usage & Strategies in Familiar Services



How Marketing Strategies are Implemented in our Daily Lives



What We Experience

- **Netflix:** Personalized movie recommendations based on watch history.
- **Amazon:** Targeted offers based on browsing and purchase history.
- **Airbnb:** Tailored travel recommendations and experiences based on user profiles and previous bookings.
- **Starbucks:** Personalized rewards and promotions based on customer transaction data.

Customers who viewed items in your browsing history also viewed

Page

Collaborative Filtering

Item	Price	Discount	Shipping
2030 半導体の地政学 (増補版) 戦略物資を… 太田泰彦 ★★★★☆ 59 Tankobon Softcover	¥1,090	60pt (3%)	Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon
半導体戦争 世界最重要テクノロジーをめぐる国… タクリス・ミラー ★★★★☆ 490 Tankobon Softcover	¥2,070	90pt (3%)	Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon
2040年 半導体の未来 小堀 隆信 ★★★★☆ 44 Tankobon Hardcover	¥1,980	198pt (10%)	Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon
観光“末”立国〜ニッポンの現状〜 (扶桑社新書) 永谷 亜矢子 ★★★★☆ 11 Paperback Shinsho	¥900	10pt (1%)	Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon
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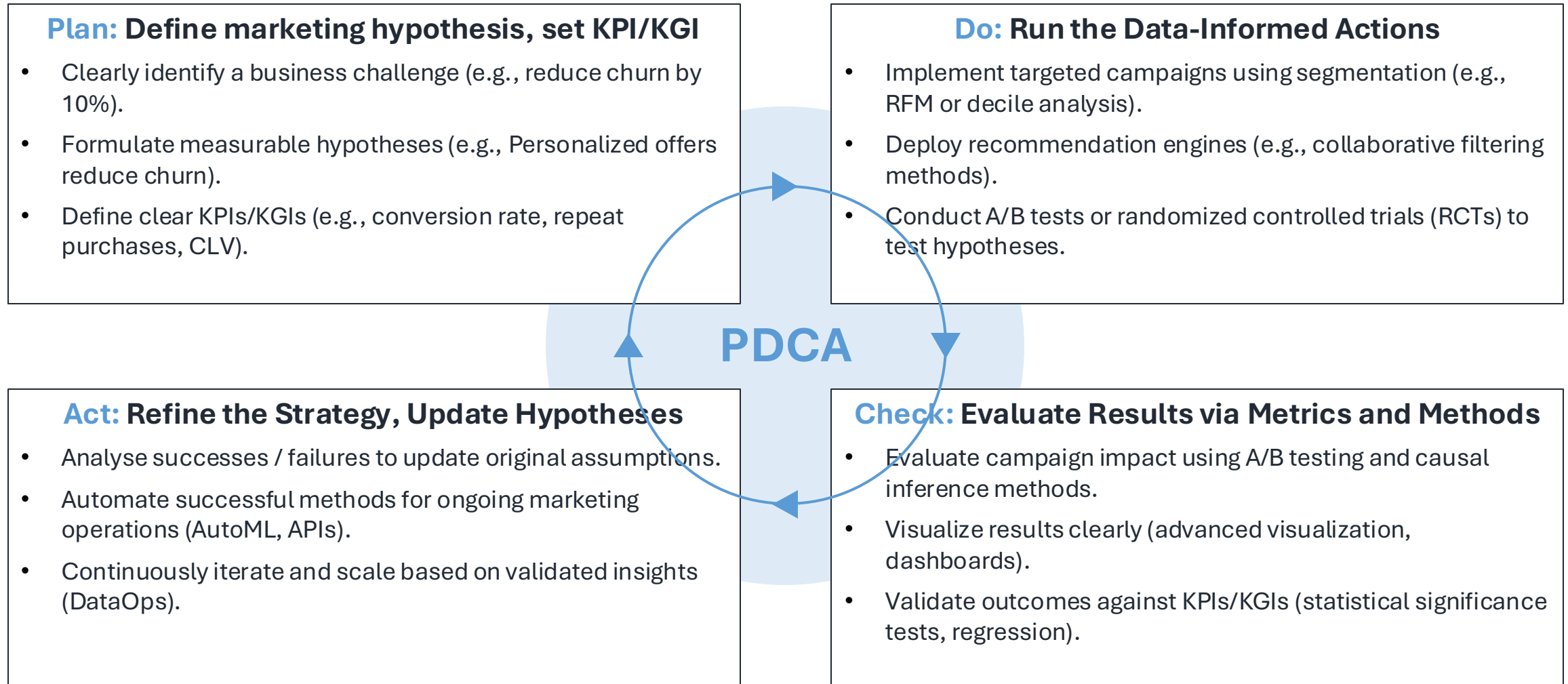
Content-based Filtering

Item	Price	Discount	Shipping
TP-Link UB400 Bluetooth USB Adapter for PC / Nano Size / Ver4.0 ★★★★☆ 89,872 -15% ¥990 List: ¥1,168 Get it as soon as Tuesday, Mar 11 FREE Shipping on orders over ¥3,500 shipped by Amazon	¥990		
Sennheiser MOMENTUM 4 Wireless Headphones, High Performance… ★★★★☆ 8,158 -18% ¥50,166 List: ¥69,960 502pt (1%) Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon	¥50,166		
Anker Soundcore Space Q45, Bluetooth 5.3 Wireless Headphone, Up to 65… ★★★★☆ 2,739 -14,990 3148pt (21%) Get it as soon as Tuesday, Mar 11 FREE Shipping by Amazon	¥14,990		
Sony WF-1000XM5 Wireless Noise Cancelling Earbuds, MISA CM Model/Fully Wireless/Outstanding Noise Cancelling ★★★★☆ 2,104 ¥32,480 FREE Shipping	¥32,480		
Amazon Gift Card Charge Type ★★★★☆ 91,950 #1 Best Seller in Gift Cards ¥100	¥100		



Data Science in Marketing

Hypothesis Driven Analysis, Data-Informed Action, and Constructive Evaluation through the PDCA Cycle



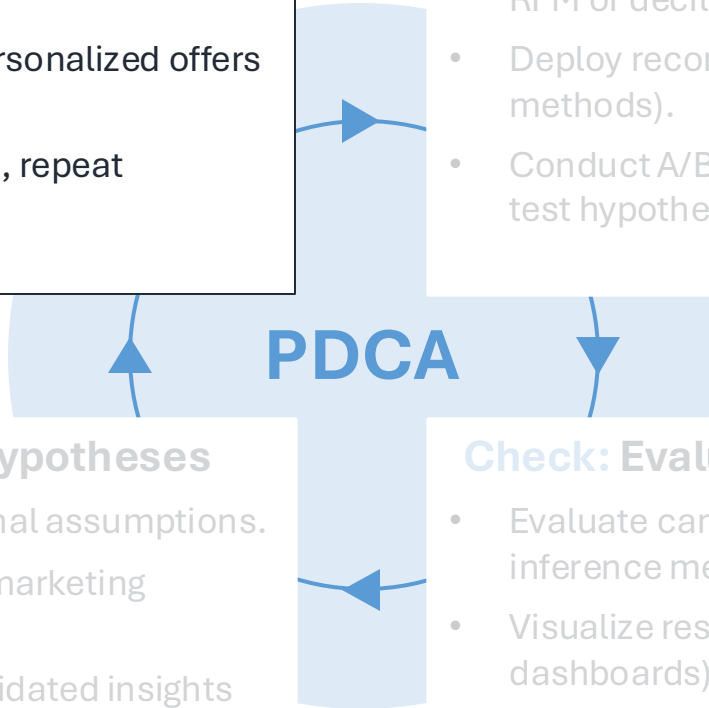
Plan: Define marketing hypothesis, set KPI/KGI

- Clearly identify a business challenge (e.g., reduce churn by 10%).
- Formulate measurable hypotheses (e.g., Personalized offers reduce churn).
- Define clear KPIs/KGIs (e.g., conversion rate, repeat purchases, CLV).

Do: Run the Data-Informed Actions

- Implement targeted campaigns using segmentation (e.g., RFM or decile analysis).
- Deploy recommendation engines (e.g., collaborative filtering methods).
- Conduct A/B tests or randomized controlled trials (RCTs) to test hypotheses.

PDCA

A large, light blue cross-shaped diagram with the text 'PDCA' in the center. Four blue arrows point clockwise from the center towards the four quadrants: up, right, down, and left.

Act: Refine the Strategy, Update Hypotheses

- Analyse successes / failures to update original assumptions.
- Automate successful methods for ongoing marketing operations (AutoML, APIs).
- Continuously iterate and scale based on validated insights (DataOps).

Check: Evaluate Results via Metrics and Methods

- Evaluate campaign impact using A/B testing and causal inference methods.
- Visualize results clearly (advanced visualization, dashboards).
- Validate outcomes against KPIs/KGIs (statistical significance tests, regression).

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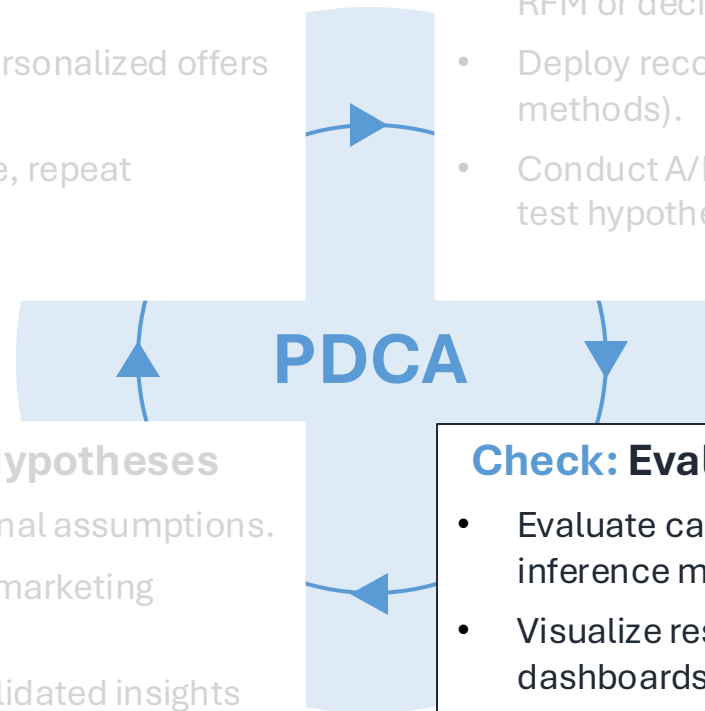
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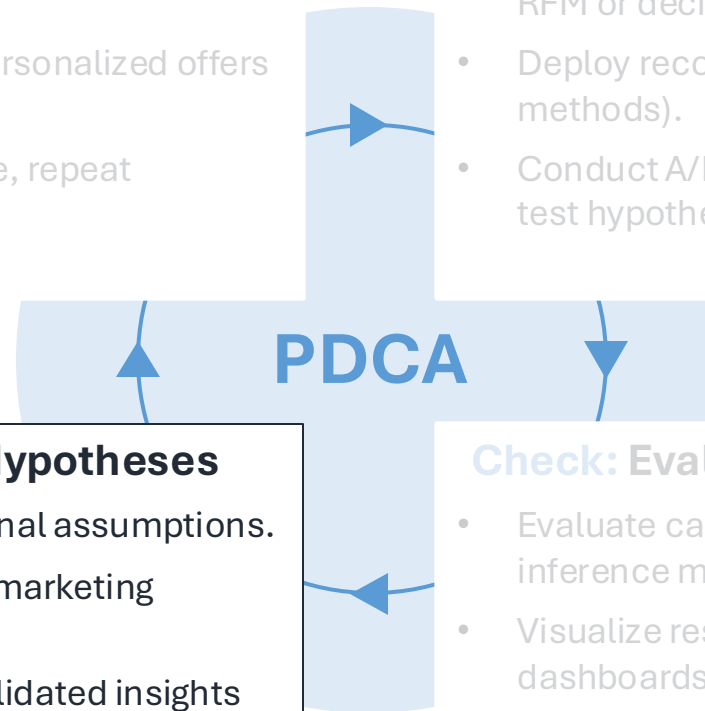
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Case



Case: Video Streaming Service “Hongo Stream”



Streaming Service Provider “Hongo Stream” (Imaginary) is Your Client and You are the Data Scientist

01

Client Requirements – Minutes of Meeting with the Chief Marketing Officer

1. Decline in **average monthly viewing time and rise in churn rate** is an urgent issue
2. The company wants to leverage **existing user data** (viewing histories, ratings, and preferences) to address these issues, but **they are not in a clean form**
3. “Hongo Stream” needs a scalable, automated solution that **adapts to user preferences without extensive manual input**
4. Clear reporting and visualization tools are required to **regularly track improvements and justify continued investment**

02

Financial Aspect – Minutes of Meeting with the Chief Financial Officer

1. “Hongo Stream” has allocated a **budget of \$500,000 USD** for developing and deploying the data-driven solution
2. Current **annual subscription revenue per user is \$150**, with a total user base of **10 million subscribers**
3. With an annual **churn rate of 15%**, “Hongo Stream” faces potential annual revenue losses of around **\$225 million**

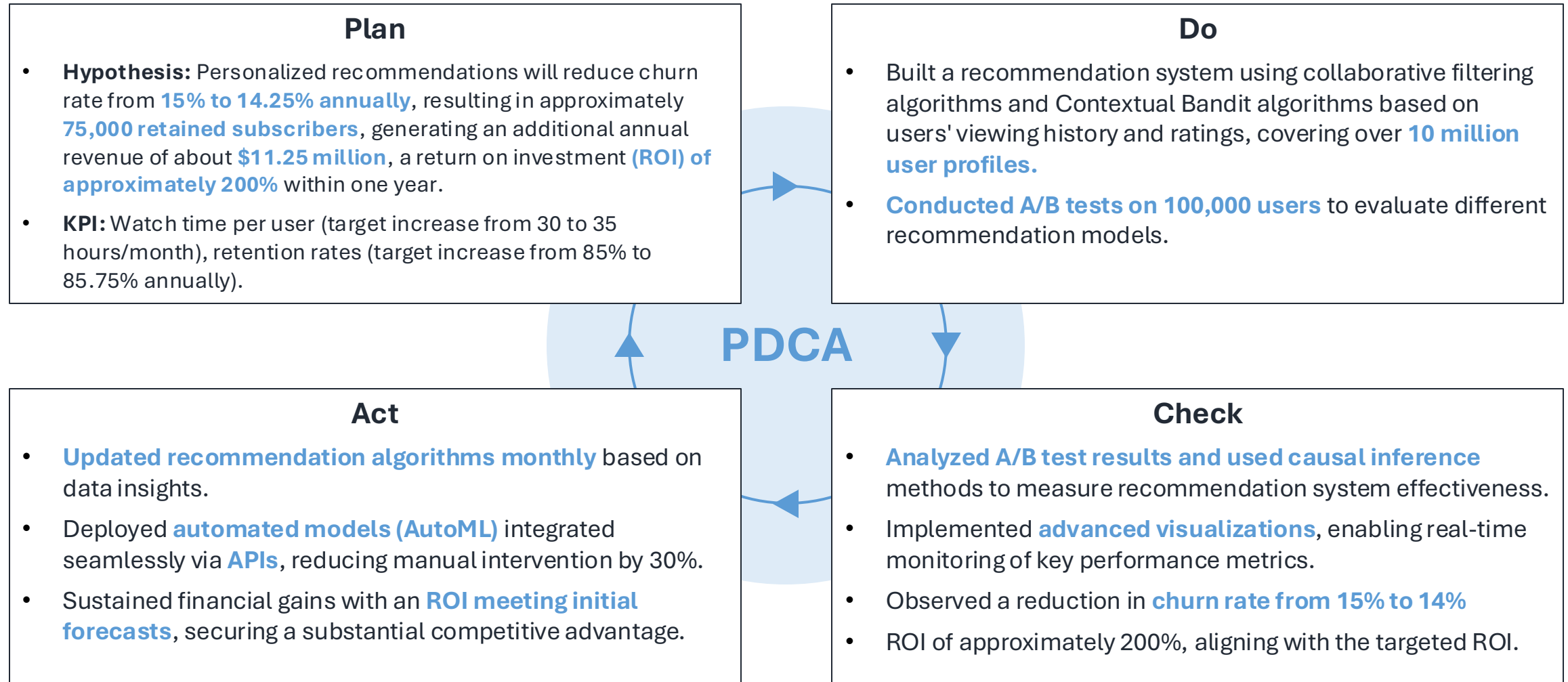
A. What would your proposal be?

B. How would you realize that proposal?

Case: Video Streaming Service “Hongo Stream”



Fundamentals are always the same



Case: Video Streaming Service “Hongo Stream”



Fundamentals are always the same

Plan

- **Hypothesis:** Personalized recommendations will reduce churn rate from **15% to 14.25% annually**, resulting in approximately **75,000 retained subscribers**, generating an additional annual revenue of about **\$11.25 million**, a return on investment (ROI) of **approximately 200%** within one year.
- **KPI:** Watch time per user (target increase from 30 to 35 hours/month), retention rates (target increase from 85% to 85.75% annually).

Do

- Built a recommendation system using collaborative filtering algorithms and Contextual Bandit algorithms based on users' viewing history and ratings, covering over **10 million user profiles**.
- **Conducted A/B tests on 100,000 users** to evaluate different recommendation models.

PDCA

Act

- **Updated recommendation algorithms monthly** based on data insights.
- Deployed **automated models (AutoML)** integrated seamlessly via **APIs**, reducing manual intervention by 30%.
- Sustained financial gains with an **ROI meeting initial forecasts**, securing a substantial competitive advantage.

Check

- **Analyzed A/B test results and used causal inference** methods to measure recommendation system effectiveness.
- Implemented **advanced visualizations**, enabling real-time monitoring of key performance metrics.
- Observed a reduction in **churn rate from 15% to 14%**
- ROI of approximately 200%, aligning with the targeted ROI.

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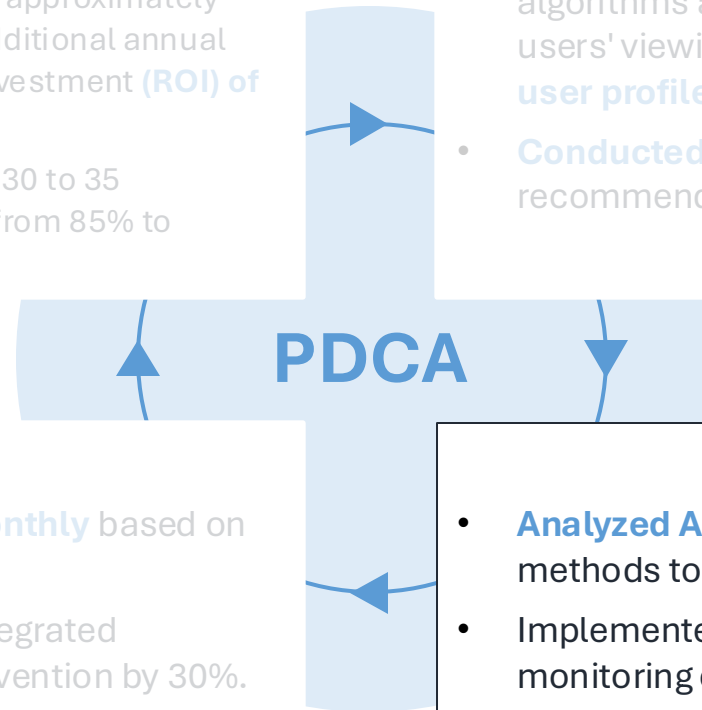
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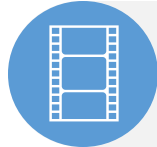


Hands-on Marketing Analysis

(Designed for Self-study)

02

Jupyter Notebook contents for this lecture



Introduction to Consumer Analysis and CRM in Marketing

- Marketing strategy
- Data exploration
- Decile analysis
- RFM analysis
- Separation analysis
- Factorization

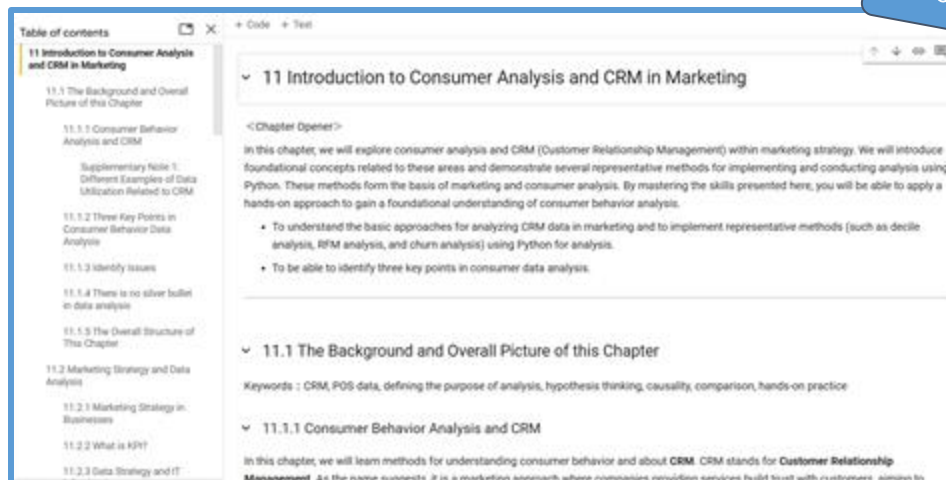


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- 11 Introduction to Consumer Analysis and CRM in Marketing
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 - 11.1.1 Consumer Behavior Analysis and CRM
 - Supplementary Note 1: Different Examples of Data Utilization Related to CRM
 - 11.1.2 Three Key Points in Consumer Behavior Data Analysis
 - 11.1.3 Identify issues
 - 11.1.4 There is no silver bullet in data analysis
 - 11.1.5 The Overall Structure of This Chapter
 - 11.2 Marketing Strategy and Data Analysis
 - 11.2.1 Marketing Strategy in Businesses
 - 11.2.2 What is KPI?
 - 11.2.3 Data Strategy and IT

11 Introduction to Consumer Analysis and CRM in Marketing

<Chapter Opener>

In this chapter, we will explore consumer analysis and CRM (Customer Relationship Management) within marketing strategy. We will introduce foundational concepts related to these areas and demonstrate several representative methods for implementing and conducting analysis using Python. These methods form the basis of marketing and consumer analysis. By mastering the skills presented here, you will be able to apply a hands-on approach to gain a foundational understanding of consumer behavior analysis.

- To understand the basic approaches for analyzing CRM data in marketing and to implement representative methods (such as decile analysis, RFM analysis, and churn analysis) using Python for analysis.
- To be able to identify three key points in consumer data analysis.

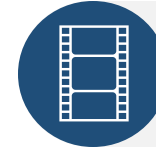
11.1 The Background and Overall Picture of this Chapter

Keywords : CRM, POS data, defining the purpose of analysis, hypothesis thinking, causality, comparison, hands-on practice

11.1.1 Consumer Behavior Analysis and CRM

In this chapter, we will learn methods for understanding consumer behavior and about CRM. CRM stands for Customer Relationship Management. As the name suggests, it is a marketing approach where companies provide services built trust with customers, aiming to

Introduction



Advanced Consumer Analysis and CRM in Marketing

- Recommendation
- Causal inference
- Automation through advanced visualization, APIs, etc.

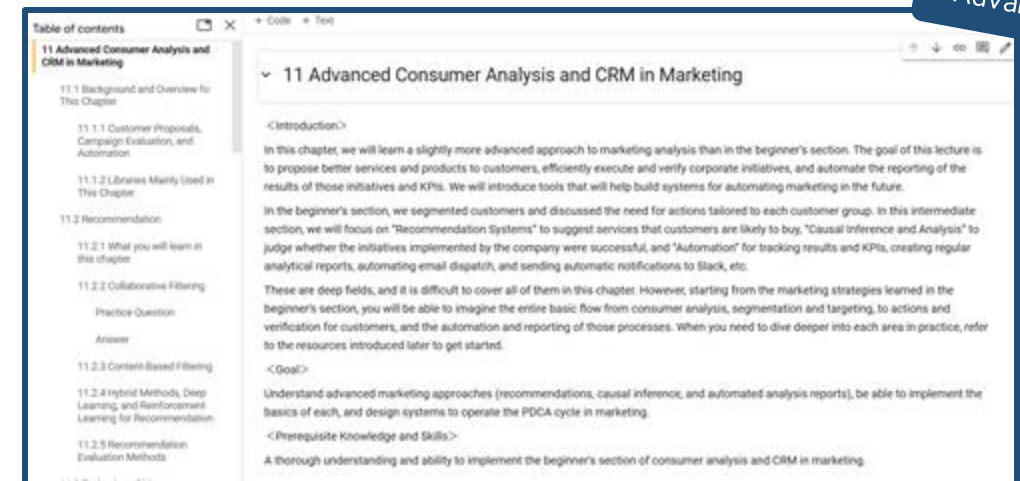


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 - 11.1.1 Customer Proposals, Campaign Evaluation, and Automation
 - 11.1.2 Libraries Mainly Used in This Chapter
 - 11.2 Recommendation
 - 11.2.1 What you will learn in this chapter
 - 11.2.2 Collaborative Filtering
 - Practice Question
 - Answer
 - 11.2.3 Content-Based Filtering
 - 11.2.4 Hybrid Methods, Deep Learning, and Reinforcement Learning for Recommendation
 - 11.2.5 Recommendation Evaluation Methods

11 Advanced Consumer Analysis and CRM in Marketing

<Introduction>

In this chapter, we will learn a slightly more advanced approach to marketing analysis than in the beginner's section. The goal of this lecture is to propose better services and products to customers, efficiently execute and verify corporate initiatives, and automate the reporting of the results of those initiatives and KPIs. We will introduce tools that will help build systems for automating marketing in the future.

In the beginner's section, we segmented customers and discussed the need for actions tailored to each customer group. In this intermediate section, we will focus on "Recommendation Systems" to suggest services that customers are likely to buy, "Causal Inference and Analysis" to judge whether the initiatives implemented by the company were successful, and "Automation" for tracking results and KPIs, creating regular analytical reports, automating email dispatch, and sending automatic notifications to Slack, etc.

These are deep fields, and it is difficult to cover all of them in this chapter. However, starting from the marketing strategies learned in the beginner's section, you will be able to imagine the entire basic flow from consumer analysis, segmentation and targeting, to actions and verification for customers, and the automation and reporting of those processes. When you need to dive deeper into each area in practice, refer to the resources introduced later to get started.

<Goal>

Understand advanced marketing approaches (recommendations, causal inference, and automated analysis reports), be able to implement the basics of each, and design systems to operate the PDCA cycle in marketing.

<Prerequisite Knowledge and Skills>

A thorough understanding and ability to implement the beginner's section of consumer analysis and CRM in marketing.

Advanced

The Next Steps

03



01



Continue learning

- Participate in Kaggle competitions and actively study successful case studies.
- Enroll in advanced courses on deep learning and emerging technologies (e.g., no-code tools, cloud services).
- Expand your international knowledge by strengthening language skills, particularly English.

02



Experimental Development

- Develop software or analytical systems to build practical experience.
- Experiment with creating web applications, social media integrations, drone and robotics operations, automation solutions, video editing projects, game development, or financial trading platforms.

03



Go into the field

- Engage in part-time or internship roles involving data analysis, system development, etc.
- Obtain hands-on project experience to solidify your technical and practical skills.
- Continuously acquire new competencies and knowledge directly applicable to your professional tasks.



Starting Your Own...?

Many Companies Are Struggling



01

Lack of Data Strategy & In-house Capability

02

Complex and Bizarre Databases

03

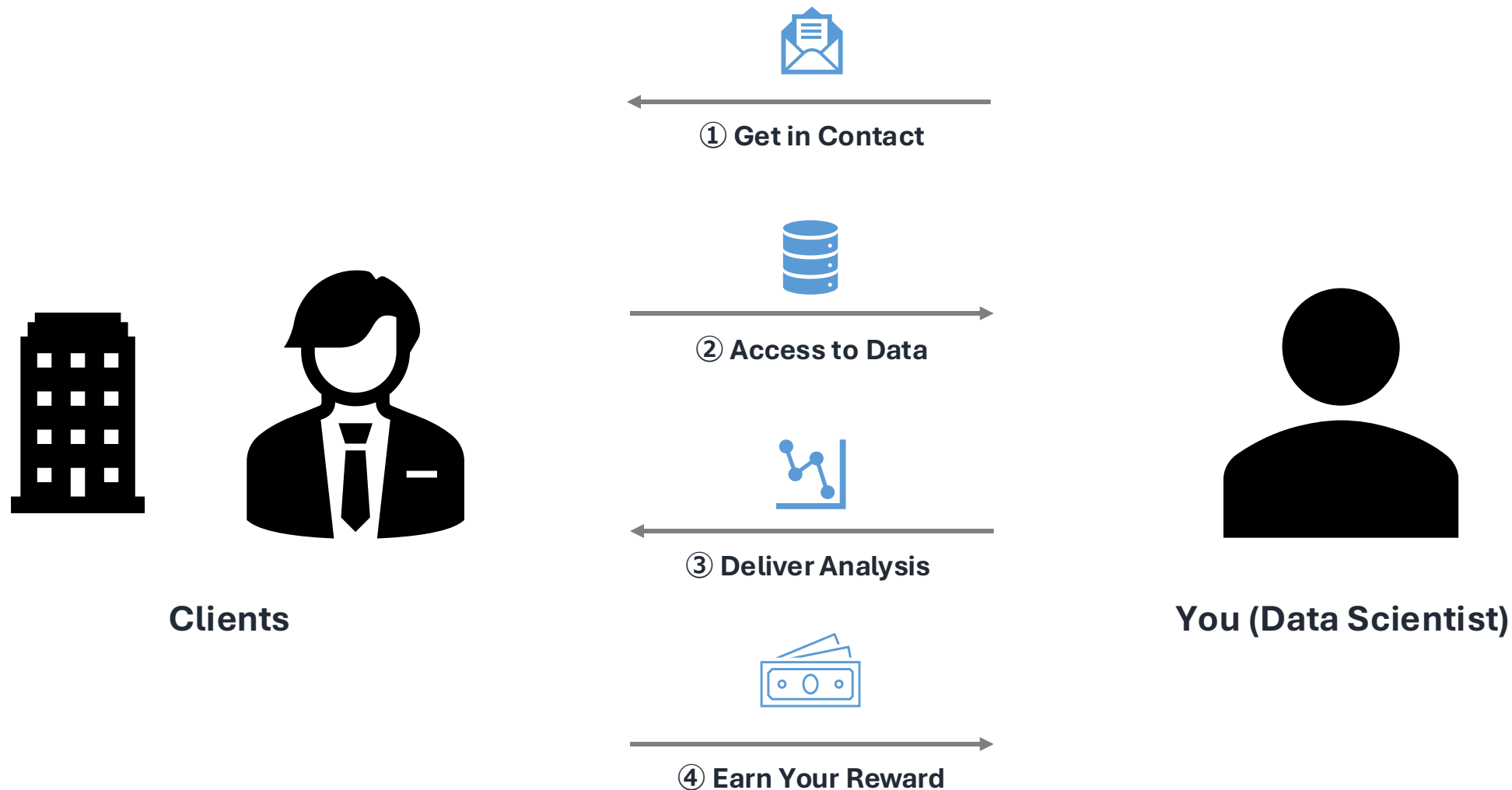
Radical Transformation in Technology, Particularly in AI and Data Science

The **ample time** and the **proximity to cutting-edge technology** are invaluable.
With the right skills, you can contribute meaningful value from **day one!**

The First Step is Always the Biggest Step



You can start small — either on your own or with a team.



Once You Get the Hang of It, Found One



Matsuo Lab Supports Entrepreneurial Journeys



① Get Ready

- Take GCI and related courses
- Get sufficient knowledge and skills around data science / AI etc.



② Build a Team

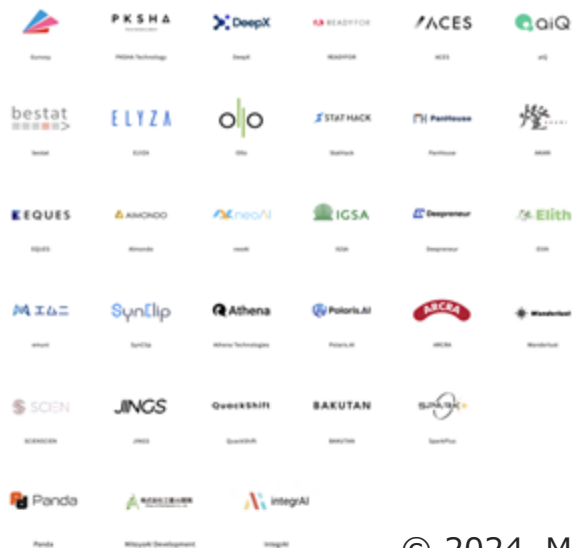
- Find Co-founders
- Find first few clients
- Deliver and Build Reputation



③ Found your Company

- Build an Organization
- Continue the Cycle of Sales, Delivery and Recruitment
- Build your Own Product

List of Matsuo Lab Startup® 32 companies (as of Mar. 2025)



IPO 2 companies

PKSHA
TECHNOLOGY

Sep, 2017



Apr, 2015

M&A 3 companies

ELYZA

Mar, 2024

STAT HACK

Jun, 2024

Awarded Forbes JAPAN 30 Under 30

燈
2022



neoAI
2024



01

Introduction to Marketing & Data Science

- Marketing is at the **core of many business operations** and serves as an ideal starting point for data science
- Fundamentals of marketing: **Marketing funnels, CRM systems, and everyday examples**
- **Hypothesis driven data strategies** and the continuous PDCA (Plan, Do, Check, Act) Cycle

02

Hands-on Marketing Analysis (Designed for Self-study)

- **Demonstration using Jupyter Notebook**
- **Data exploration, decile analysis, RFM analysis, etc.**
- **Recommendation engine, causal inference, etc.**

03

The Next Steps

- **From Theory to Practice: [Getting Into the Field](#)**
- **How Matsuo Lab Has Produced [Numerous Startups and the Model Behind It](#)**

